

Liquid State Machines for Real-Time Neural Simulations

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Abstract

We propose a new, large artificial neural network in simulation of a system processing visual signals. Our model is built of 4880 Hodgkin-Huxley neural cells, which are organized into Retina (Input) built on a 28×28 square-shaped grid that is divided into four (2×2) patches, and a modular Cortex of Liquid State Machine (LSM) model with four computational columns, each on a $8 \times 8 \times 16$ grid attached to every retina patch. We simulated 1 second of biological work of the visual system and measured that in the current experimental setup it required 285 CPU seconds. We also observed different spiking patterns within each LSM column, and measured them using euclidean distance. Such behaviour is typical for LSMs. We also explored Apache Spark analytics platform for an in-memory processing of the readout signal. Our approach is scalable and could be considered in some expert-devices with the ability to classify geometrically different and time variable patterns. Our research can lead to a better understanding of pattern recognition phenomena in the human brain, as well as the development of machine vision and creation of new intelligent devices using LSMs.

Keywords— liquid computing, liquid state machine, pattern recognition, signal processing, visual system, computer vision

Physics Subject Headings— spiking neurons, pattern formation, neuronal network activity, artificial neural networks

1 Introduction

Our brain consists of approximately a hundred billion neurons. The data provided by different authors have led to a broad range of 75–125 billion neurons in the whole brain [15]. Neurons are organised in microcircuits, also known as neural columns. They differ by purpose in the human brain [8].

Excitable media, as introduced by [11] in the theory of synchronous concurrent algorithms, provide a great framework for computations. A new approach to microcircuit computing was suggested by Maass [16]. In Maass’s Liquid State Machine (LSM) theory, the brain, or its fragments, are treated as a liquid. The model provides an alternative to the Turing machine [19]. Moreover, a mathematical analysis shows that there are in principle no computational limitations of liquid state machines [16].

Neural networks are a great tool for modelling different systems, including complex biological or technological systems. Artificial neural networks facilitate biomedical signal processing, biomedical data analysis and interpretation, as well as models for the analysis of system behaviour, including the prognosis of results of selected activities [20].

Several successful applications of the LSM framework have been delivered in the area of artificial neural networks, or to solve engineering tasks such as the design of nonlinear controllers

[18]. Moreover, we know that cortical microcircuits seem to be very useful for computing on perturbations [24].

Direct observation of neuronal activity of individual neurons is not possible given the current state of art in human neuroimaging [1]. To achieve brain-scale simulations and to investigate emergent properties of brain circuits we need better building blocks to simulate the whole neural circuits with higher fidelity.

There has been an increased development in the performance and capability of neural simulations in the past years. As a result the simulator and supercomputer technology have now developed to the point that the actual process of setting up and simulating a realistic neural model is now practical [9].

In spite of that some key challenges remain. One of them is that the number of synaptic connections in the neural models offering higher fidelity have shown to exceed the current memory capacity of hardware that is available to researchers [14]. The other major challenge to the simulation of neuronal networks is handling both the computations and data generated by the large number of synaptic inputs to a single neuron, and scaling these neurons to the larger structures [13].

This paper presents a simulation framework for spiking neural network simulation and provides a simple, extensible retina-LGN-cortex model. We also leverage a novel distributed simulation setup, with in-memory processing of the readout signal. We also extend the prior work of [23], in building a higher fidelity bio-inspired visual system resembling mammalian visual cortex. The new model has a more complex retina, allowing to process input patterns generated by viewing a resolution of a MINST dataset [6] in near real-time. We achieve simulation results from four LSM columns, and calculate the Euclidean distance of states in each of the four columns of the system to illustrate the differences in spiking patterns.

We present a model that helps in understanding the signal processing phenomena within each column, and is extensible for further research on signal processing in multiple hypercolumns of the brain.

1.1 Liquid State Machines

In contrast to common computational models, Maass’s Liquid State Machine introduced in 2002 [16] does not require information to be stored in stable states of a computational system. In the same year a Liquid Computer term was coined as a novel strategy for real-time computing on time series [18]. Such a computer consists of the four main components:

- Input $i(\cdot)$, a continuous input stream.
- Liquid column, or a filter L^N that maps the input into function $Y^N(t)$:

$$Y^N(t) = (L^N i)(t). \quad (1)$$

- Memory-less readout map m^N .
- Output $o(t)$:

$$o(t) = m^N(Y^N(t)). \quad (2)$$

In Maass’s LSM architecture described in [16] the function of time series $i(\cdot)$ is injected as input into the liquid column L^N , creating at time t the liquid state $Y^N(t)$, which is transformed by a memory-less readout map m^N to generate an output $o(t)$.

Maass introduced two macroscopic properties of LSM, a Separation Property (SP) and an Approximation Property (AP), that can provide improvements with respect to classification applications. SP measures the dispersion between projected liquid states from different classes, whereas the AP indicates the concentration of the liquid states that belong to the same class.

The architecture is under active development. Recently [22] proposed to use a group of locally connected LSM reservoirs to form an ensemble of liquids. This approach could simulate higher levels of connectivity in LSMs that are in close proximity, and lower levels of connectivity for these that are spatially further apart.

1.2 Real-time Processing

We propose to combine the LSM architecture with Apache Spark [27]. Apache Spark can be used as a distributed, fault tolerant data processing engine for extracting insights at scale with near-real time speeds [28]. It is an open source computing framework that unifies streaming, batch, and interactive big data workloads. We use it to improve the analysis of outputs generated by neural columns simulated with GENESIS [3] programming framework.

The fundamental building block of Spark architecture is Resilient Distributed Dataset (RDD). RDDs are in-memory objects on which all the operations on Spark platform are performed, and it happens in a distributed way [26].

An RDD is a collection of entities, similar to rows or records. RDD functionality makes the distributed processing possible by allowing to split the data it contains across all the data nodes of a Spark cluster. RDDs are immutable; once created they cannot be edited, updated, or appended to. They are considered resilient because they tolerate node failures within the cluster, and can be reconstructed in case of a node failure [26, 29].

From the programming perspective they are similar to Java collections, but under the Spark layer they are partitioned and distributed across multiple computing nodes. Unlike in the case of Hadoop and HDFS, Spark RDDs represent the data in-memory for each of the machines in the cluster. The distribution of the data across the computational nodes allows Spark to process the data in parallel. Several processes can be run on an individual subset of data by a cluster [29].

RDDs can be mutated (be appended or changed). There are only two operations that are permitted on an RDD: [28]:

- Transformation (resulting in creation of a new RDD, with all the edits that are needed).
- Action (or request for a result, which causes Spark to execute a set of transformations defined for a dataset).

Spark follows lazy evaluation of operations by keeping a record of the series of transformations requested by the user on the dataset. It groups the transformation in an efficient way when an action is requested. This allows an RDD to be reconstructed even if the node it lives on crashes. RDD can be created when a file is read, or a transformation of another RDD is called. Each RDD keeps metadata in which it keeps track of where it came from. This feature is called the RDD lineage, and it allows an RDD to reconstruct itself through re-performing all the recorded transformations [26].

Since its beginnings, the platform offers machine learning libraries. Spark 1.x provides support for ML with *spark.mllib*. Spark 2.x, the current version works with *spark.ml* and it offers an entirely new set of APIs for developers, which can work with data frames, and not directly with Resilient Distributed Datasets. The other advantage of the platform for machine learning is that Spark offers libraries for hyper-parameter tuning, allowing to choose the best model for a given use case [17]. Moreover, the recent developments in the Spark project increased the execution speed on the platform between 10 and 100 times [29].

2 Materials and Methods

2.1 Signal Processing Model

We propose a bio-inspired model of a visual system that consists of two main modules. Our approach is in line with the LSM architecture proposed by Maass [16]. There are two main components of our model:

1. Input (Retina, as presented in Fig. 1).
2. Liquid (Cortex, built of four identical LSM columns, each as presented in Fig. 2).

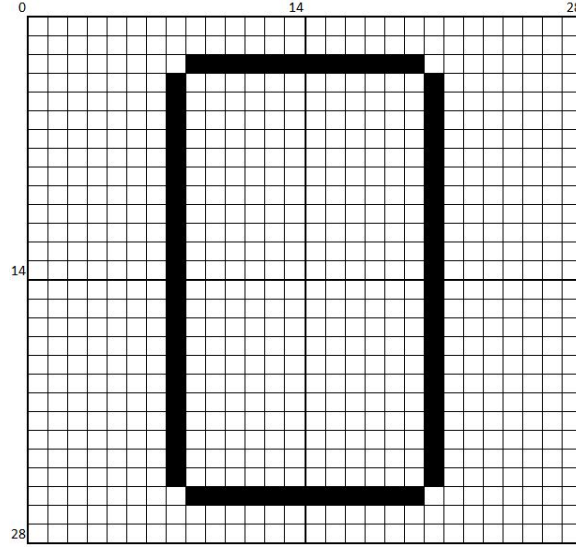


Figure 1: Retina of the proposed visual system with the stimulating patterns (Input).

All the simulations discussed in this paper were programmed using GEneral NEural SIMulation System (GENESIS) [4]. All the neurons used in the simulations were built according to the Hodgkin-Huxley model [10].

Similar to the old model presented in [23], our new Hodgkin-Huxley Liquid State Machine (HHLSM) model uses the same high fidelity multi-compartmental neurons. The soma of each neuron uses biologically similar voltage-activated sodium and potassium channels. We build on a well known conductance-based model describing how action potentials in neurons are initiated and propagated in electrical circuit [25]. The GENESIS parameters we used in our simulations can be organised into four groups:

- Main resistances $R_x = 0.3 \, \Omega$, $R_n = 0.33 \, \Omega$.
- Capacitance $C_n = 0.01 \, \text{F}$ and potential incl. $E_n = 0.07 \, \text{V}$, $E_k = 0.0594 \, \text{V}$ (soma compartment), $E_k = 0.07 \, \text{V}$ (dendrite).
- Conductance $G_K = 360 \, \Omega^{-1}$ and $G_{Na} = 1200 \, \Omega^{-1}$ (for each of the ionic channels).
- Physical characteristics, a soma with circular shape and diameter of $30 \, \mu$, dendrites and axon length of $100 \, \mu$.

The exact values for these parameters were achieved experimentally by [25] to provide a high biological fidelity of the model. A detailed description of the Hodgkin-Huxley model can be found in [10].

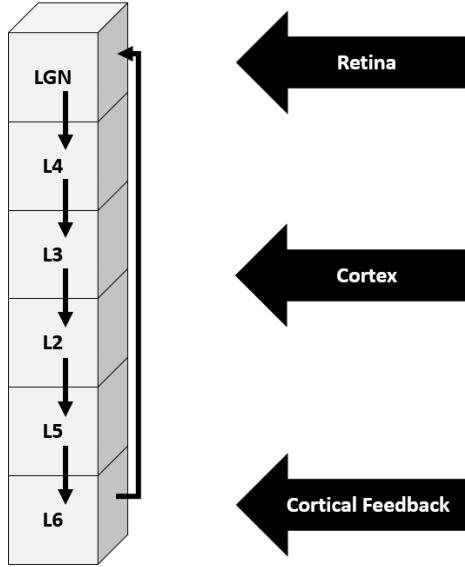


Figure 2: Structure of the LSM column, a fundamental computational microcircuit of our model (Liquid).

The Retina was built on a 28×28 square-shaped grid and divided into four patches (2×2). Each patch is connected to one of the four HHLSM columns which simulate Lateral Geniculate Nuclei (LGN), and later the ensemble of cortical microcircuits. The retinal cells are only connected to the LSM column through the LGN layer. Each HHLSM consists of 1024 neural cells placed in a cuboid of $8 \times 8 \times 16$. The structure of each column in the models is the same (Fig. 2). The model contains four columns in total that form the Liquid stimulated by Retina.

There are 90% of excitatory connections established among layers and neurons of each layer and 10% of inhibitory connections. Additionally, Layers L6 of LSM columns are connected with LGNs of other HHLSMs in the same way (i.e. with the probability of 10%), simulating the corticothalamic feedback. For simplicity, we did not implement the pathway of the possible inter-column connections. Each connection in the model is characterised with a delay parameter and random weight. We can treat the Liquid as a single hyper-column made of four independent neural columns, a part of periodic structure of the simulated cortex.

2.2 Experimental Setting

We performed all the simulations described in this paper using a simple Spark cluster with a single master node, and two worker nodes. The system was built around computing resources available on a free Google Colaboratory Cloud platform [2]. The machine was equipped with the Intel Xeon processing units running at 2.30 GHz, using 256KB L2 Cache in a single computing component, and having 2 cores per chip. The 64-bit system had 12 GB of RAM, and worked under the control of Linux (Ubuntu 18.04.5 LTS) with the kernel version of 4.19.112+. We installed the latest Spark-3.0.2 with hadoop 2.7 using OpenJDK Runtime Environment 11.0.10. The model was implemented in GENeral NEural SIMulation System GENESIS v.2.4¹. Both

¹GENESIS v.2.4 <https://github.com/dbeeman/genesis-2.4beta-files>.

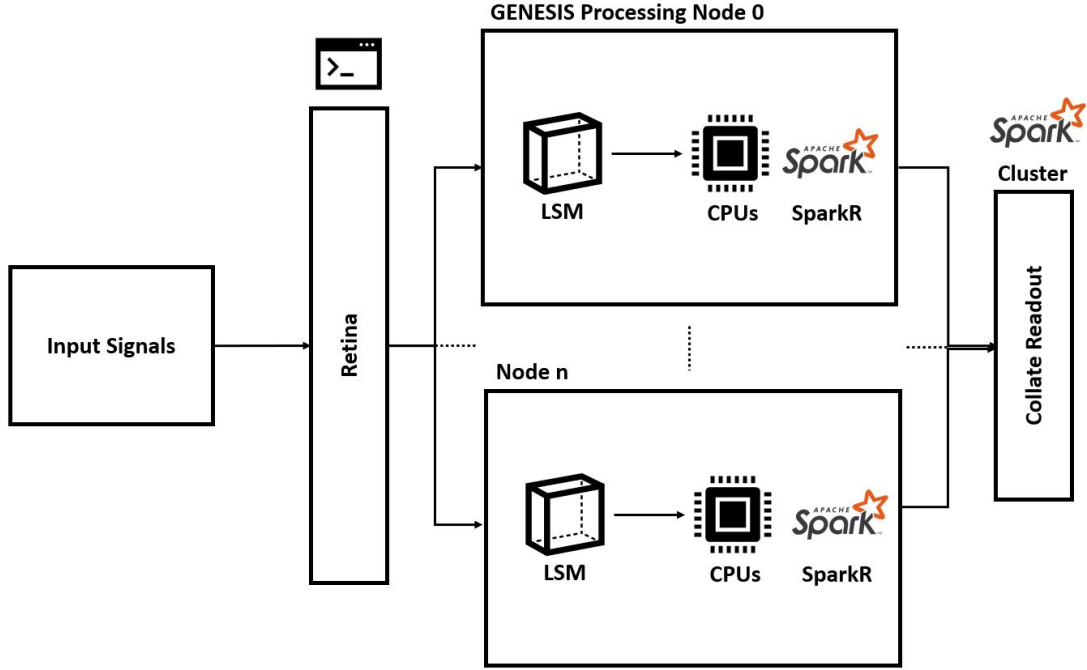


Figure 3: High level simulation setup including GENESIS and Spark components of the framework.

results and source code for the simulations are available on GitHub repository². Fig. 3 presents a diagram with simulation setup, including GENESIS and Spark components, along-side their relative roles in the overall approach.

3 Results

We created a system consisting of 784 retina cells and a Liquid State Machine organised into 4 computational columns, 1024 cells each. In contrast to the original Maass’s LSM using integrate and fire neurons, our architecture is built of a more biologically realistic model of neural cells proposed by Hodgkin-Huxley.

In total our system was built using 4880 artificial neurons. They simulated a simple visual system, processing input signals through a square-shaped grid of 28x28 pixels. We stimulated the retinal cells with a pattern resembling the digit of 0. The input signal (Fig. 1) was encoded in the Liquid state, and a unique pattern of spikes was observed in each of four LSM columns, as visualised in Fig. 6 and 7, and measured in Table 1.

Fig. 4 and 5 present two views of the spiking response of our retina grid. Retinal cells chosen for each pattern were stimulated with random spike trains with the average rate of 200 Hz. The spikes marked in black were triggered by input signals simulating the pattern of ‘shorter part’ of the 0 shape (2 * 12 stimulating spike trains), whereas the spikes marked in blue were generated by the ‘longer parts’ of the 0 shape stimulating our retina model (2 * 22 stimulating signals). This is important to observe, because it shows that our retina response generates expected signals when stimulated, and these signals can be passed to the subsequent parts of the system,

²LiquidComputer <https://github.com/KarolChlasta/LiquidComputer>.

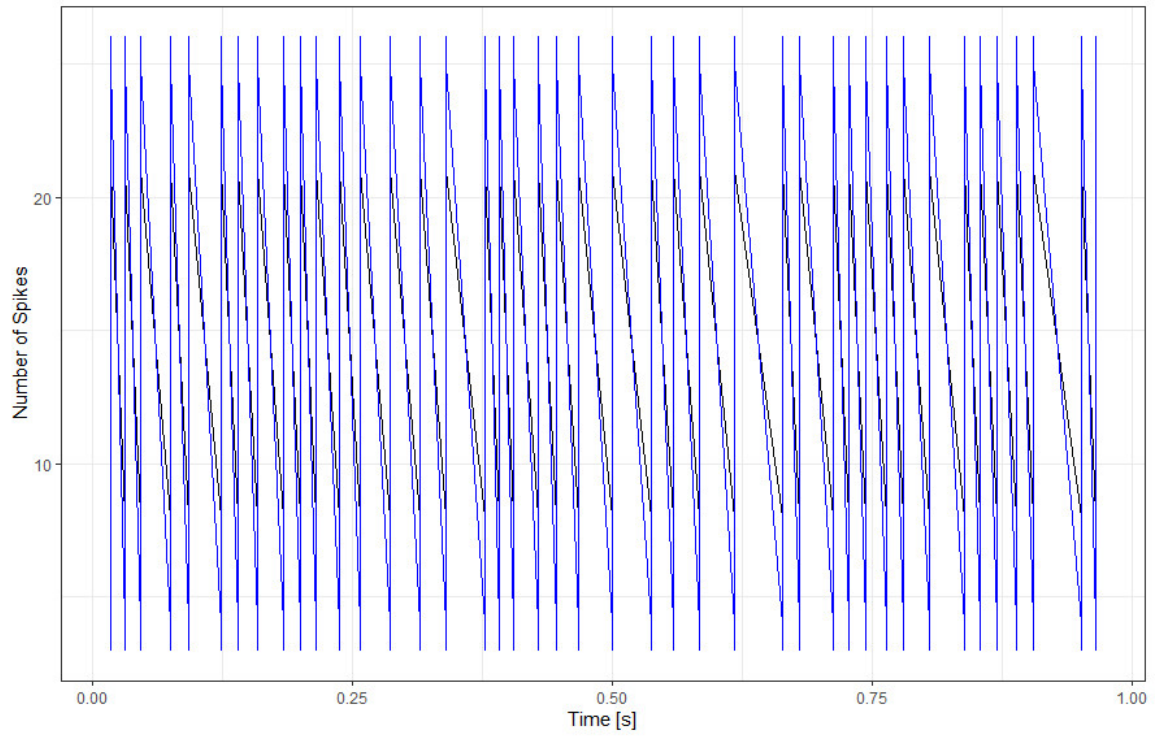


Figure 4: Number of spikes generated by Retina (Input) stimulation for a 2D (28x28) setup within the simulation time of 1 second.

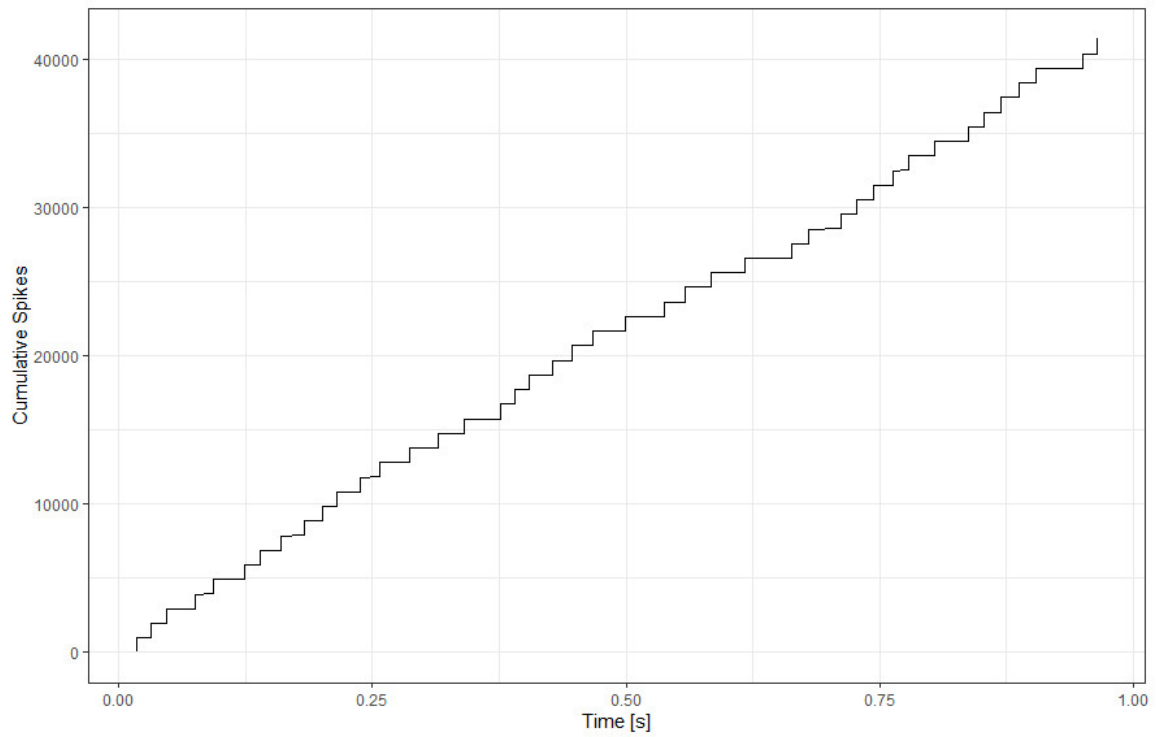


Figure 5: Cumulative number of Retina spikes in each ms of simulation reaching 40000 before the end of simulation time of 1 second.

Table 1: Euclidean distance between vectors of states of all four LSM columns

Euclidean Distance	Column 1	Column 2	Column 3	Column 4
Column 1	0	3.822376	4.308967	4.375152
Column 2	3.822376	0	1.556321	7.617872
Column 3	4.308967	1.556321	0	8.062106
Column 4	4.375152	7.617872	8.062106	0

as presented in Fig. 2.

Fig. 6 and 7 present cumulative spiking activity for each of the LSM columns. We performed a 2D kernel density estimation to visualise the structure of cumulative spikes with contour lines. We observe different dynamics of these spikes generated within each LSM column. Such behaviour is typical for LSMs, and can be measured for each column, using their vectors of states.

Finally, Table 1 presents the matrix of euclidean distances between four vectors of states of each LSM column during the time of experiment. This different pattern is an important feature for the downstream processing of visual information because we confirm the liquid computing abilities of neural microcircuits. We observe that the spiking pattern of each of the four columns is slightly different, what is numerically expressed in the distance values calculated for each pair of columns.

We simulated a single second of this simple visual system in 20000 steps. We measured that in the current setup of the system (the simulation time step of 0.00005 second) that a single second of simulation required 285 CPU seconds (4:45 min). Each LMS column generated on average 1.34 MB of spiking data, with retina adding additional 113 KB.

Apart from gathering the spiking times for each neuron, our system can also measure cell membrane potential for each neural cell in the simulation. In this approach, the amount of data generated increases approximately by 165 times, so the additional stream generated in this way reaches 1 GB per second. We have not noticed any unexpected changes of the data stream over the time of simulations.

4 Discussion

In this paper we demonstrated a proof-of-concept for a liquid state machine, as a bio-inspired computational technique in simulation of a visual system. We report the spike numbers for a complex artificial neural network built of a single 2D retina, and four 3D LSM columns. We simulated 4880 Hodgkin-Huxley neurons organised in layers resembling cortical microcircuits. Application of liquid computing ideas [16] allowed to decrease the number of neurons in the model constructed.

Our results prove that the proposed architecture based on four HHLSMs columns reacts to a visual stimulus in an expected manner, with each of the four LSM columns exhibiting a slightly different spiking pattern. We simulated a single second of the visual system, and that single second required required 285 CPU seconds.

The key limitation of our approach was lack of computational resources to perform longer, and more complex simulation. In future we would also like to use the parallel version of GENESIS to spread the processing across all available CPUs. We believe that the simulation time can be significantly shortened by using a more powerful computational cluster. We expect that our multi-column LSM model can be scaled into a parallel multiprocessor simulation, as the current architecture allows a single column to be simulated on a single processor for optimal performance.

Our model is flexible. The Retina can be formed of other shapes. A single column can be

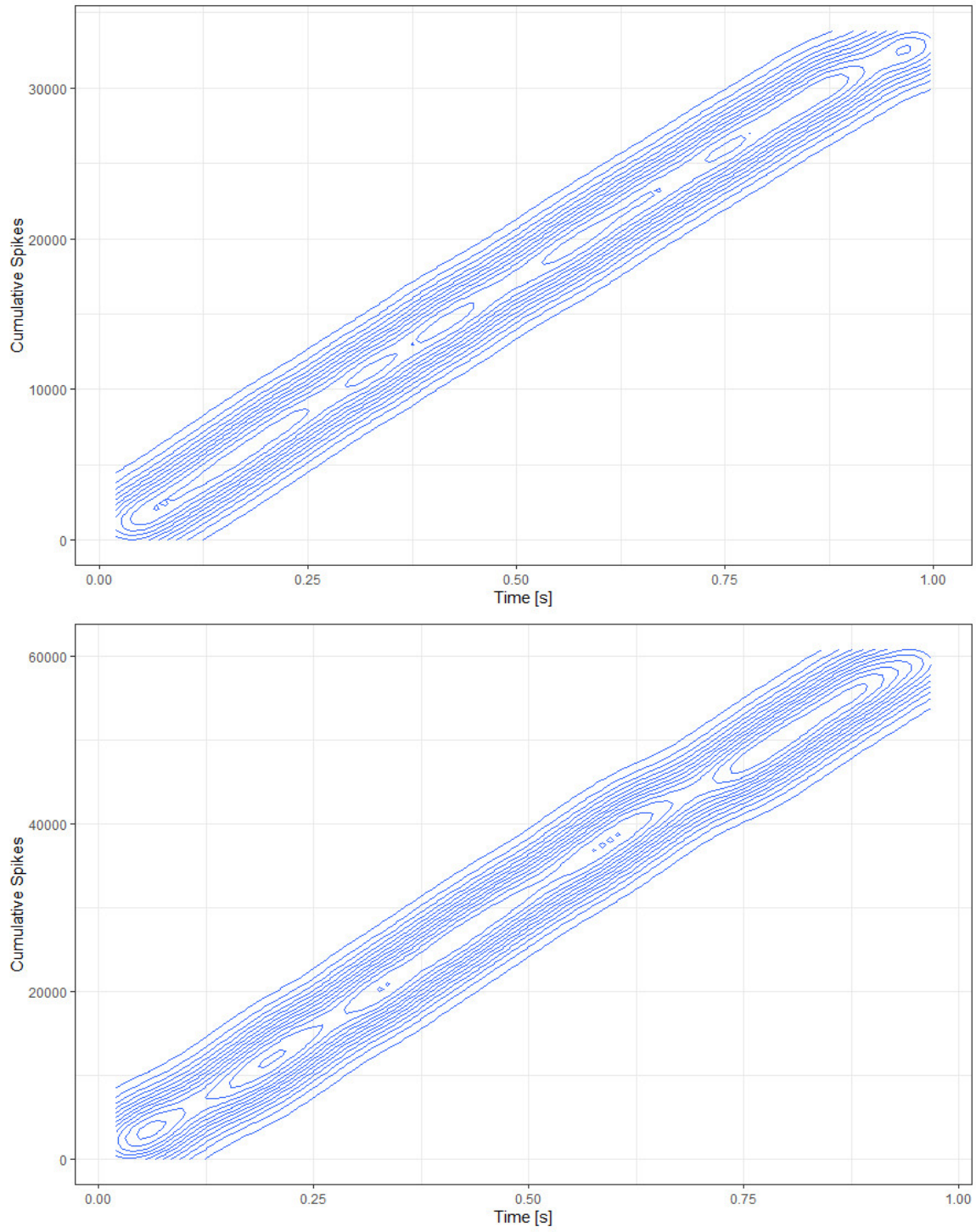


Figure 6: Visual signal processing in the first and second LSM columns of our system.

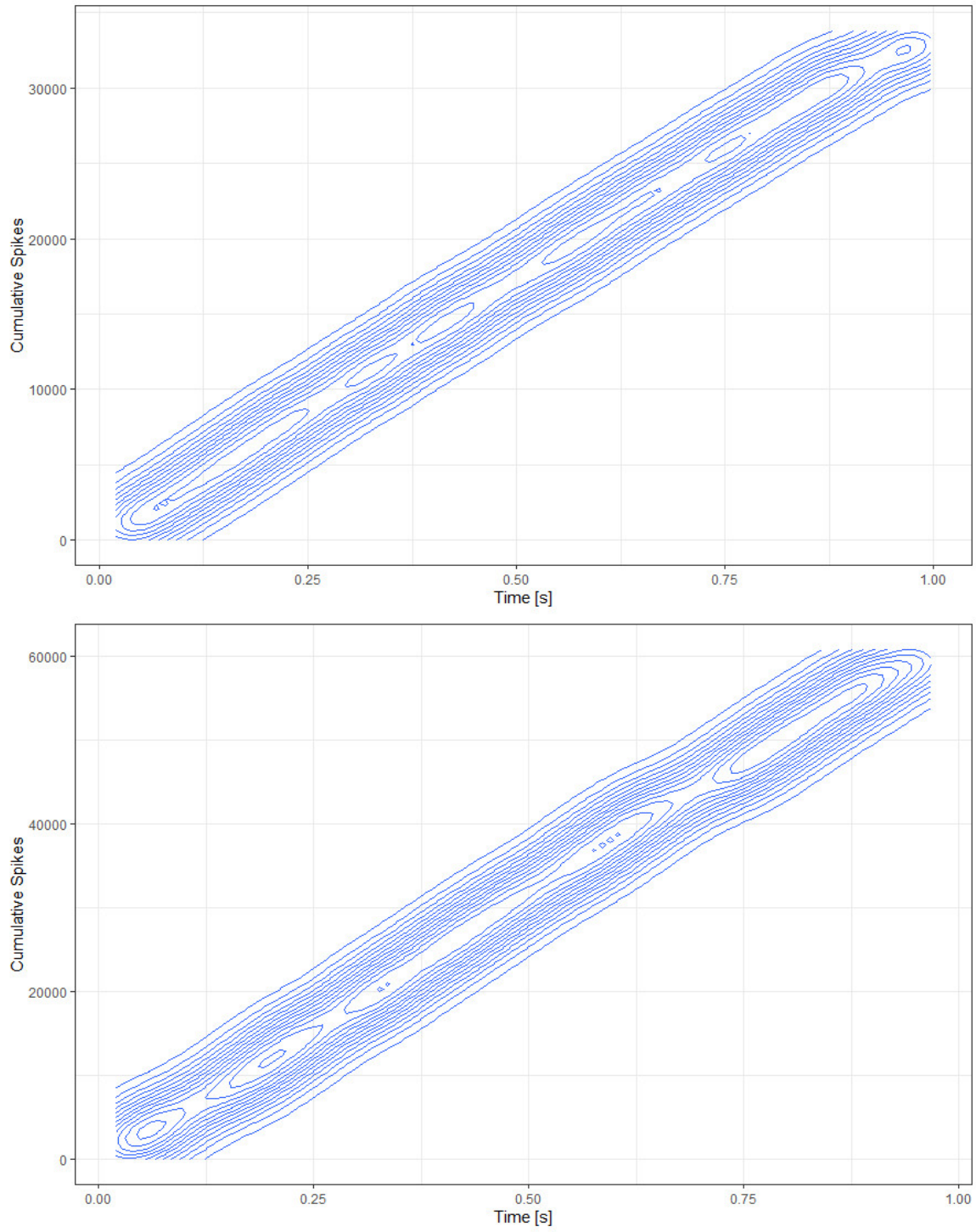


Figure 7: Visual signal processing in the third and fourth LSM columns of our system.

connected to a single retinal cell, or a patch of 4 (2×2), 16 (4×4), 256 (16×16), or more cells, depending on the required stimulus size, and the number of CPUs available. For the proposed retina size (28×28), if each retina cell is connected to a single corresponding HHLSM column, the model can conduct a simulation of about 802 thousand Hodgkin-Huxley neural cells. The current size of retina allows processing of input patterns generated by viewing a MINST [6] dataset, which is a very popular baseline among computer vision researchers.

Apart from neuronal simulations, another possible application of our system could be in automating the operations of the industrial systems, such as the one described in [12]. While evaluating the system, the authors [21] used Electroencephalography (EEG) measurements with EMOTIV EPOC+ 14-Channel Wireless Headset. They reported that the highest increase in mental involvement of the human operator happened during the data supplementation phase, where the value of the investigated mental load reached 90%. Our system could reduce the degree of the operator’s mental involvement by providing an intelligent ‘robotic eye’. The topology of the container mentioned in [12] allows the system to use our LSM model without any modifications of the retina.

Authors believe that much more realistic models could be developed on the basis of the modelling work presented in this paper. One of the directions of further development of the model proposed could be to include the lateral inhibitory connections between the LSM columns. Inhibitory connections are very common in the neocortex, and lateral inhibition has been shown to play an important role in sharpening the distinctions between similar inputs [5]. Additionally, lateral inhibition plays an important role in population coding, stabilising the mean field potential, and greatly improving the ability of a group of neurons to accurately represent a given stimulus by creating negatively correlated firing patterns [7]. As such, this feature could enhance pattern recognition capabilities of the system, and it is worth including in next version of the model.

These models could be analysed in real time by streaming readout to the Apache Spark cluster, and pushed out to designated databases or live dashboards.

5 Conclusions

To conclude, the proposed LSM architecture coupled with Apache Spark allows for the analysis and visualisation of results for an ever increasing number of simulated neurons, potentially reaching even several millions of neural cells. Adopting the proposed building blocks extends the toolbox of neuroinformatics. It can lead to the creation of even more sophisticated, higher fidelity models and open new possibilities to observe dynamics of such a visual system in real time. Following Maass’s [16] ideas and applying a readout for liquid state analysis we can also imagine some new expert-devices able to classify geometrically different and time variable patterns. Therefore, implementing neural models and arranging a proper LSM architecture of the simulated cortex can lead to a better understanding of both machine vision and pattern recognition phenomena taking place in the real brains.

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